

Syllabus

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Course Title

Algorithms for DNA Sequencing

Course Instructors

Ben Langmead and Jacob Pritt

Course Description

We will learn the concepts, algorithms and data structures that are foundational to the analysis of DNA sequencing data. We will use what we learn, together with our programming skills, to implement these and similar ideas. We will work with real data: real genome sequences and real sequencing data.

To succeed in this course, it is helpful to have some undergraduate-level background in computer science, though we have worked hard to make the course accessible even to students with minimal or no computer science background. You definitely *do not* have to be a trained biologist. You don't have to be an expert on genomics and genomes. You don't have to know how a DNA sequencer works. You just have to be curious about these things, and to be open to learning a little about genomics and sequencing as you go.

You will need Python programming skills. All of the code that we'll show you in class, in the lecture notes and in the practical sessions, will be written in Python. We will sometimes ask you to modify or adapt our examples written in Python. So if you're not so confident in your Python skills, you might want to get some more practice before you enroll, or you might go ahead and enroll, but if you do you should set aside some extra time to learn and practice as you go along.

Course Content

Module 1: DNA sequencing, strings and matching: Why study this? DNA sequencers and how they work. How DNA can be represented as a string. Using Python to parse and manipulate real genome sequences and real DNA sequencing data. Naive exact matching.

Module 2: Preprocessing, indexing and approximate matching: Improving on naive exact matching with Boyer-Moore. Preprocessing and indexing. Indexing: grouping and ordering. k -mers and k -mer indexes. Approximate matching and the pigeonhole principle.

Module 3: Edit distance, assembly, overlaps: Hamming and edit distance. Algorithms for computing edit distance. Dynamic programming. Global and local alignment. De novo assembly. Overlaps and overlap graphs.

Module 4: Algorithms for assembly: Shortest common superstring and the greedy version. How repetitive DNA makes assembly difficult. De Bruijn graphs and Eulerian walks. How real assemblers

work. The future of assembly. Wrap up.

Lecture Materials

Lecture videos will be released at the beginning of the course and will be available thereafter. You are welcome to view them at your convenience. Accompanying each video lecture will be a PDF copy of the slides.

Programming Homework

There are four programming homework assignments. You may begin submitting them as soon as the course opens. Programming homework 1 is due at the end of the first week, programming homework 2 is due at the end of the second week, programming homework 3 is due at the end of the third week, and programming homework 4 is due at the end of the fourth week. See the Homework page for exact due dates and hard deadlines.

Quizzes

There are four quizzes. You may begin submitting them as soon as the course opens. Similar to the programming homework scheduled, quiz 1 is due at the end of the first week, Quiz 2 is due at the end of the second week, Quiz 3 is due at the end of the third week, and Quiz 4 is due at the end of the fourth week. See the Quiz pages for exact due dates and hard deadlines.

Scoring

You may attempt each homework and quiz up to 3 times. The score from your most successful attempt will count toward your grade.

Hard deadlines and soft deadlines

The due date for each programming homework and quiz is available on the respective Homework and Quizzes page. For Programming Homework 1-3 and Quizzes 1-3, the hard deadline is five days after the due date. Each day late will incur a 10% penalty, but if you use a late day, the penalty will not be applied to that day. There is no grace period for Programming Homework 4 or for Quiz 4, and you may not use late days to extend the Programming Homework 4 or Quiz 4 due dates.

Late Days for Quizzes and Homework

You are permitted 5 total late days for Programming Homework 1-3 and Quizzes 1-3. Using a late day moves the due date forward one day so you can avoid late submission penalties. You may not use

late days for Programming Homework 4 or Quiz 4.

Grading policy

- Quiz 1 = 10%
- Quiz 2 = 10%
- Quiz 3 = 10%
- Quiz 4 = 10%
- Programming Homework 1 = 15%
- Programming Homework 2 = 15%
- Programming Homework 3 = 15%
- Programming Homework 4 = 15%

You must receive a final grade of 70% or better to pass the course. If you are enrolled in Signature Track and pass the course, you will receive a verified certificate. Signature Track participants who earn a grade of 90% or better will receive a Distinction designation on their verified certificate.

Typos

- We are prone to a typo or two - please report them and we will try to update the notes accordingly.
- In some cases, the videos may still contain typos that have been fixed in the lecture notes. The lecture notes represent the most up-to-date version of the course material.

Plagiarism

Johns Hopkins University defines plagiarism as "...taking for one's own use the words, ideas, concepts or data of another without proper attribution. Plagiarism includes both direct use or paraphrasing of the words, thoughts, or concepts of another without proper attribution." We take plagiarism very seriously, as does Johns Hopkins University.

We recognize that many students may not have a clear understanding of what plagiarism is or why it is wrong. Please see the following guide for more information on plagiarism:

http://www.jhsph.edu/academics/degree-programs/master-of-public-health/current-students/JHSPH-StudentReferencing_handbook.pdf

It is critically important that you give people/sources credit when you use their words or ideas. If you do not give proper credit -- particularly when quoting directly from a source -- you violate the trust of your fellow students.

The Coursera Honor code includes an explicit statement about plagiarism:

I will register for only one account. My answers to homework, quizzes and exams will be my own work (except for assignments that explicitly permit collaboration). I will not make solutions to homework, quizzes or exams available to anyone else. This includes both solutions written by me, as well as any official solutions provided by the course staff. I will not engage in any other activities that will dishonestly improve my results or dishonestly improve/hurt the results of others.

Technical Information

Regardless of your platform (Windows or Mac) you will need a high-speed Internet connection in order to watch the videos on the Coursera web site. It is possible to download the video files and watch them on your computer rather than stream them from Coursera and this may be preferable for some of you.

Here is some platform-specific information:

Windows

The Coursera web site seems to work best with either the Chrome or the Firefox web browsers. In particular, you may run into trouble if you use Internet Explorer. The Chrome and Firefox browsers can be downloaded from: *Chrome*: <http://www.google.com/chrome> *Firefox*: <http://www.mozilla.org>

Mac

The Coursera site appears to work well with Safari, Chrome, or Firefox, so any of these browsers should be fine.

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